THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · HARD LEVEL UNIT TEST SET 3

Chapters 1–4: **Large Numbers, Arithmetic Expressions, A Peek Beyond the Point, Expressions using Letter-Numbers** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Which of these is the greatest? 1

(a) 0.45 **(b)** $\frac{1}{2}$ **(c)** $\frac{2}{5}$ **(d)** 0.405

2. If $x = 6$, the value of $2x^2 + 5$ is: 1

(a) 79 **(b)** 78 **(c)** 77 **(d)** 76

3. The number of seconds in a day, 30 days is: 1

(a) 2,59,200 **(b)** 25,88,400 **(c)** 25,92,000 **(d)** 25,93,000

4. The value of $250 \div (5 \times (7 - 2)) + 8 \times (6 + 3)$ is: 1

(a) 82 **(b)** 81 **(c)** 83 **(d)** 84

5. $1.5 \times 0.2 \times 0.6 =$ 1

(a) 0.18 **(b)** 1.8 **(c)** 0.018 **(d)** 18

6. If $a = 4$ and $b = 5$, the value of $3a^2 - 7b$ is: 1

(a) 13 **(b)** 15 **(c)** 14 **(d)** 12

7. The number of zeros in 1,25,000 × 80 is: 1

(a) 9 **(b)** 8 **(c)** 6 **(d)** 7

8. The value of $(12 + 8)^2 - 12^2 - 8^2 - 2 \times 12 \times 8$ is: 1

(a) 12 **(b)** 8 **(c)** 1 **(d)** 0

9. How many times does 0.25 go into 6? 1

(a) 25 **(b)** 23 **(c)** 26 **(d)** 24

10. Simplify $7(x + 3) - 4(x - 2)$: 1

(a) $3x + 13$ **(b)** $3x + 29$ **(c)** $3x - 13$ **(d)** $11x + 29$

11. 1 crore = 1,00,00,000; the number of crores in 4,50,00,000 is: 1

(a) 45 **(b)** 450 **(c)** 0.45 **(d)** 4.5

12. Compare 20 × 13 and 19 × 14. The first minus the second is: 1

(a) -4 **(b)** -5 **(c)** -6 **(d)** -7

13. 0.4 of a number is 3.2. The number is: 1

(a) 8 **(b)** 10 **(c)** 7 **(d)** 9

14. The perimeter of a rectangle with sides $7x + 6$ and $2x - 3$ is: 1

(a) $28x$ **(b)** $10x + 18$ **(c)** $18x + 6$ **(d)** $9x + 3$

15. A tank holds 3,50,000 litres. How many such tanks hold 1,26,00,000 litres? 1

(a) 46 **(b)** 35 **(c)** 36 **(d)** 37

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Find the decimal number exactly halfway between 0.3 and 0.4. 2

17. Find the value of $2x^2 - 3x + 4$ when $x = 5$. 2

18. How many seconds are there in 30 days? Write it in the Indian and the international system. 2

19. Evaluate $250 \div (5 \times (7 - 2)) + 8 \times (6 + 3)$. 2

20. Find the missing number: $1.5 \times \square = 3.6$. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Find the value of $1.5 \times 0.6 + 2.4 \div 0.6$. 3

22. A rectangle has length $(7x + 3)$ cm and breadth $(2x - 1)$ cm. Find its perimeter as an expression and its value for $x = 5$. 3

23. Arrange 90,87,654, 90,78,654, 98,07,654 in descending order and find the difference between the largest and the smallest. 3

24. Evaluate 14×18 and $(14 + 3) \times (18 - 3)$ and find the difference between the two values. 3

OR

Two friends start from the same place and walk in opposite directions at 4.2 km/h and 3.8 km/h. How far apart are they after 1.5 hours? After how many hours will they be 32 km apart? 3

25. Write the sum of 4 consecutive numbers starting at n as an expression in n, and evaluate it for $n = 5$. 3

OR

The speed of light is about 3,00,000 km per second. How far does it travel in 60 seconds? Write the answer in words. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. A city has 4,20,000 houses. Each house uses 150 units of electricity per month. Find the units used by the city in one month and in a year (12 months). Write the yearly figure in words and round it to the nearest crore. 5

OR

A school fete: 50 children each buy 4 tickets at ₹12 each. 30 adults buy 3 tickets at ₹20 each. The organisers spend ₹700 on prizes and ₹500 on decoration. Write one expression for their profit and evaluate it. 5

27. A tank of capacity 84 litres is filled by a pipe at 9.5 litres per minute, but a leak drains 2.5 litres per minute. Find the net gain per minute and the time to fill the empty tank. 5

OR

In a pattern, figure n has $5n + 2$ tiles. Find 5

(i) the tiles in figure 12,

(ii) the total tiles in figures 1 and 2, and

(iii) how many more tiles figure 12 has than figure 11.

END OF QUESTION PAPER **Total: 50 Marks**

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